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Lee et al.

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(54) **ANTENNA STRUCTURE AND DISPLAY DEVICE INCLUDING THE SAME**

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H01Q 1/24 (2006.01)
H01Q 1/36 (2006.01)

(52) **U.S. Cl.**

CPC ***H01Q 21/08*** (2013.01); ***H01Q 1/243*** (2013.01); ***H01Q 1/36*** (2013.01)

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CPC H01Q 21/08; H01Q 1/22; H01Q 1/243; H01Q 1/38; H01Q 1/36; H01Q 21/065; H01Q 9/0407; H01Q 1/24; H01Q 1/2291; H01Q 1/44; H01Q 1/46; H01Q 1/50; H01Q 21/28; H01Q 21/30; G01S 13/62; G01S 7/03

See application file for complete search history.

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(57) **ABSTRACT**

An antenna structure includes a first radiator group including a plurality of first radiators arranged in a first direction, a second radiator group including a plurality of second radiators arranged in a second direction perpendicular to the first direction, first transmission lines connected to each of the first radiators at the same layer as that of the first radiators, and second transmission lines connected to each of the second radiators at the same layer as that of the second radiators.

16 Claims, 8 Drawing Sheets

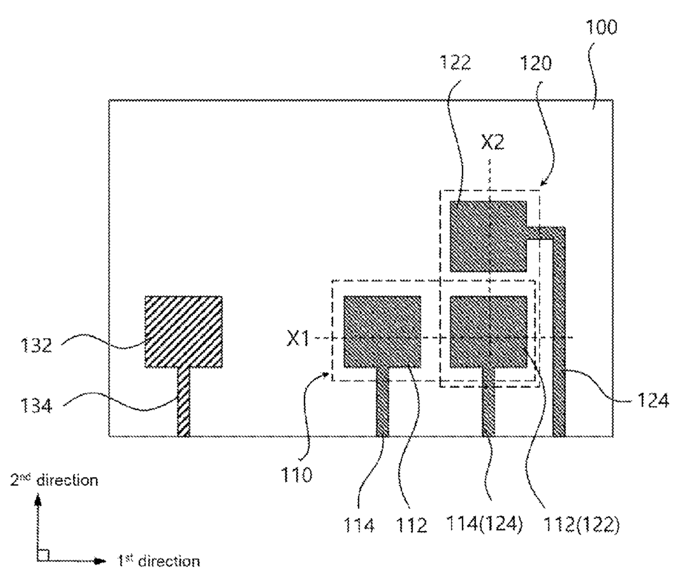


FIG. 1

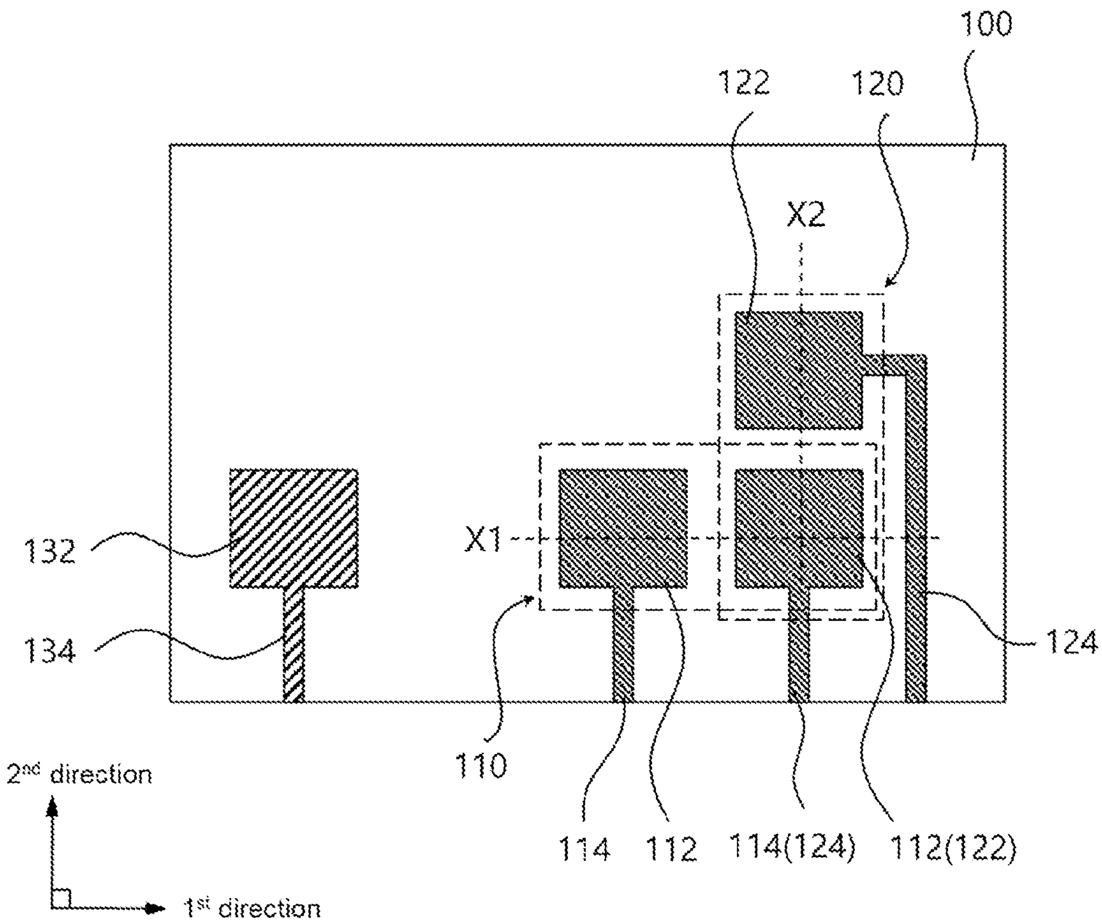

100
122
120
X2
X1
132
134
124
110
114
112
114(124)
112(122)
2nd direction
1st direction

FIG. 2

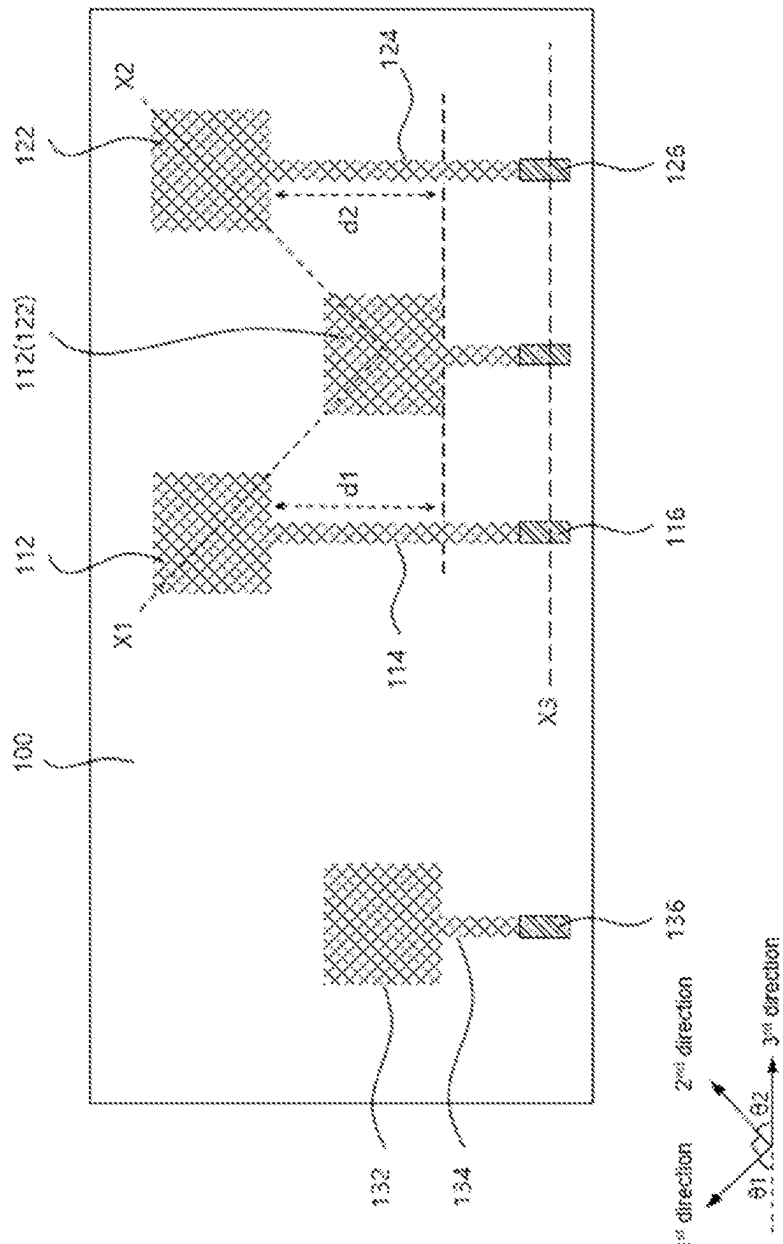

100
112
112(122)
122
X1
X2
X3
114
116
124
126
d1
d2
132
134
136
1st direction
2nd direction
3rd direction
θ1
θ2

FIG. 3

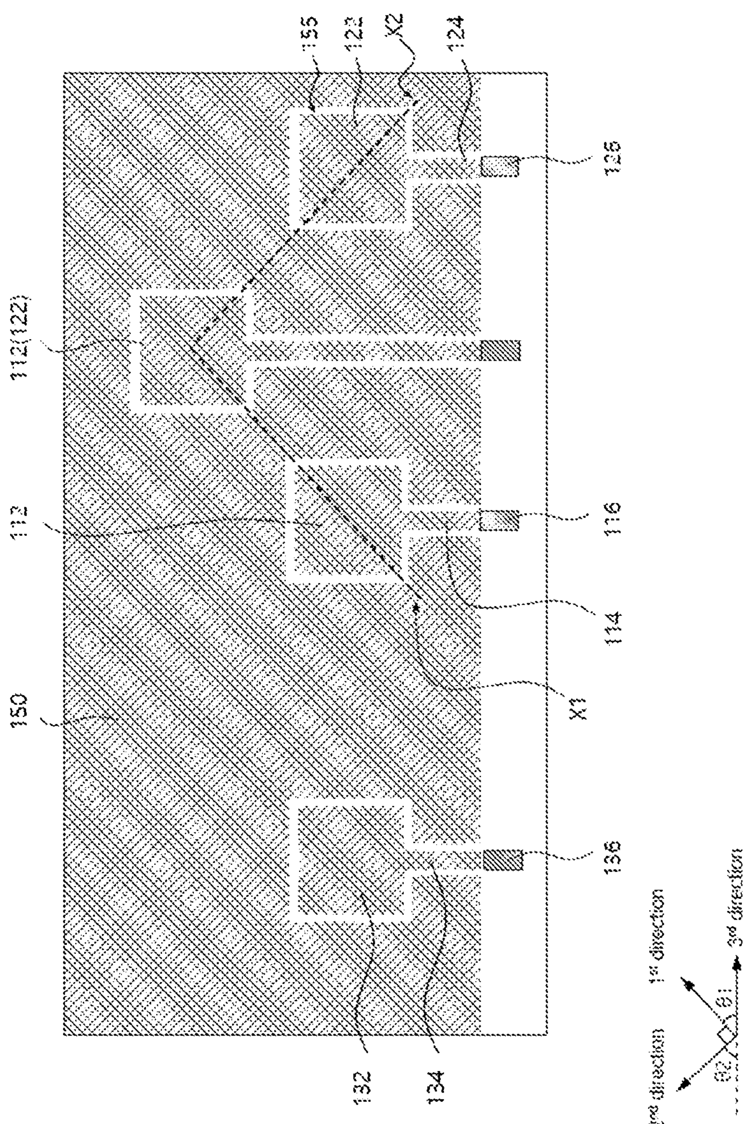

155
122
X2
124
126
112(122)
112
116
114
X1
150
136
132
134
1st direction
2nd direction
3rd direction
θ1
θ2

FIG. 4

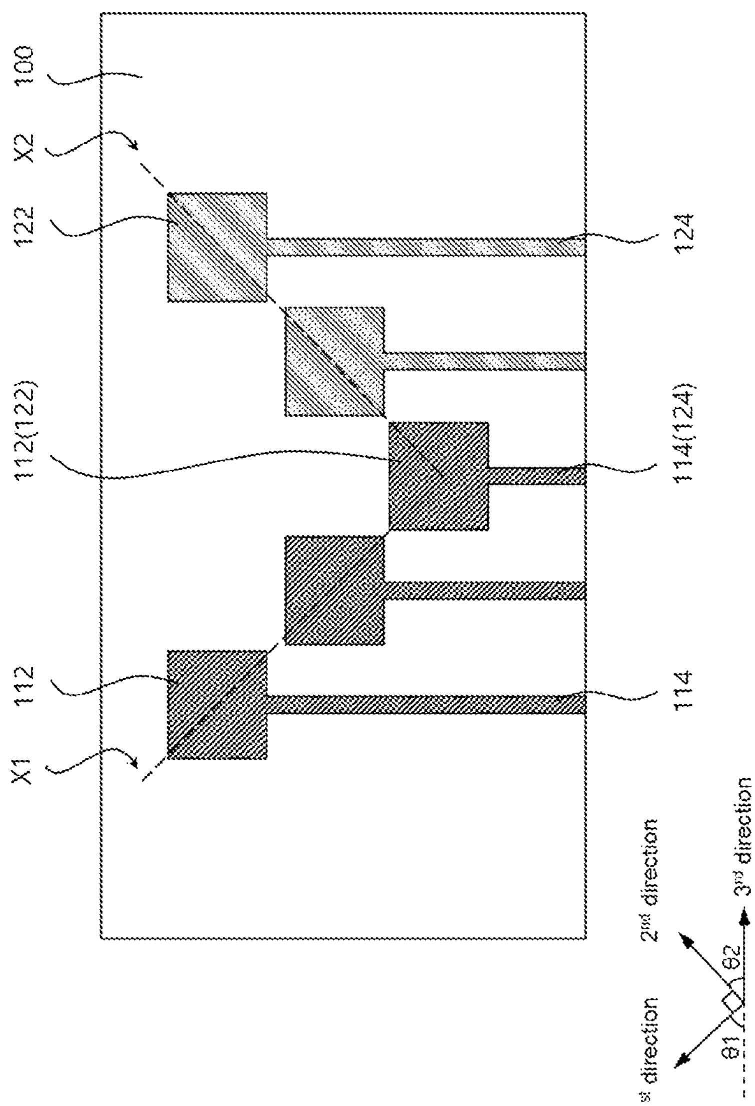

100
X2
122
124
112(122)
114(124)
112
114
X1
1st direction
2nd direction
3rd direction
θ1
θ2

FIG. 5

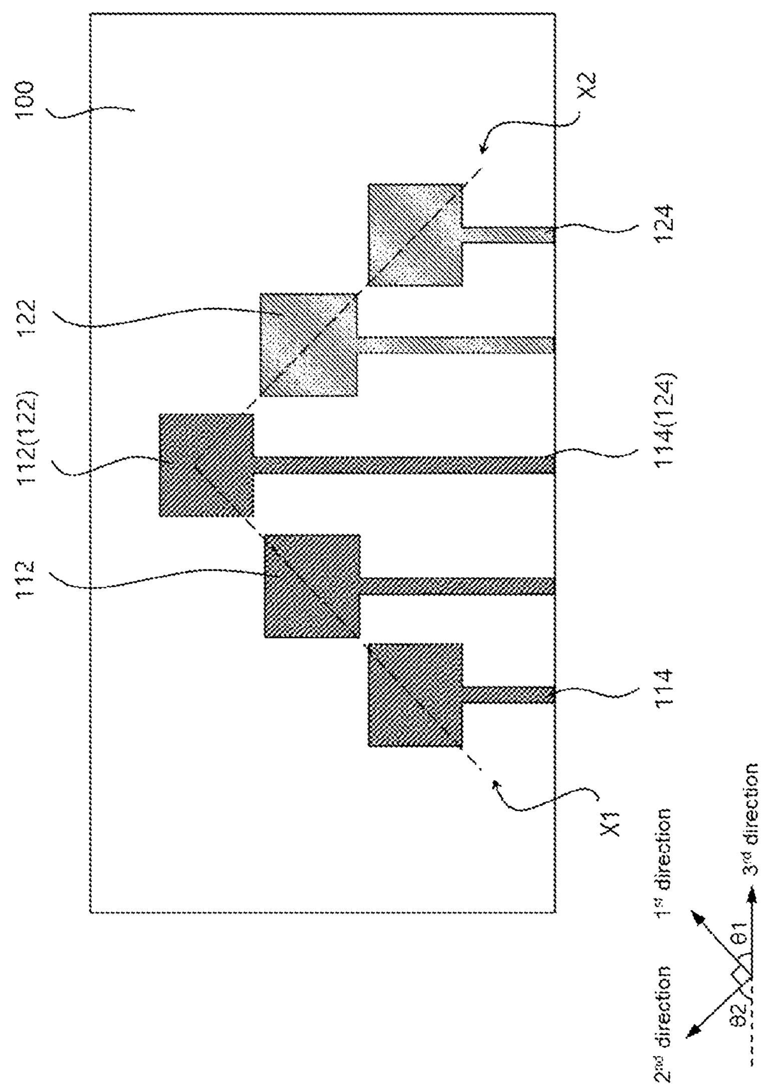

100
122
112(122)
112
X2
124
114(124)
114
X1
1st direction
2nd direction
3rd direction
θ1
θ2

FIG. 6

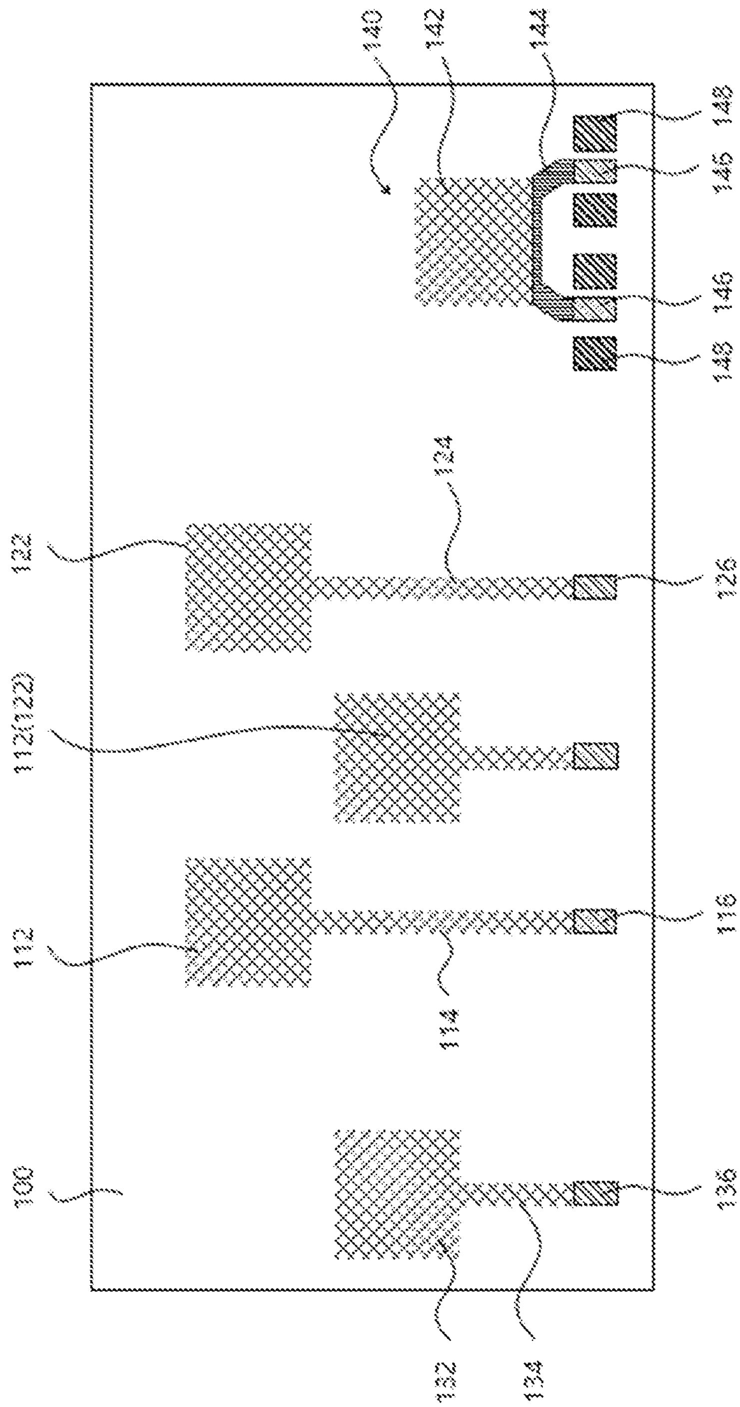

140
142
144
148
145
146
148
122
124
126
112(122)
112
114
116
100
136
132
134

FIG. 7

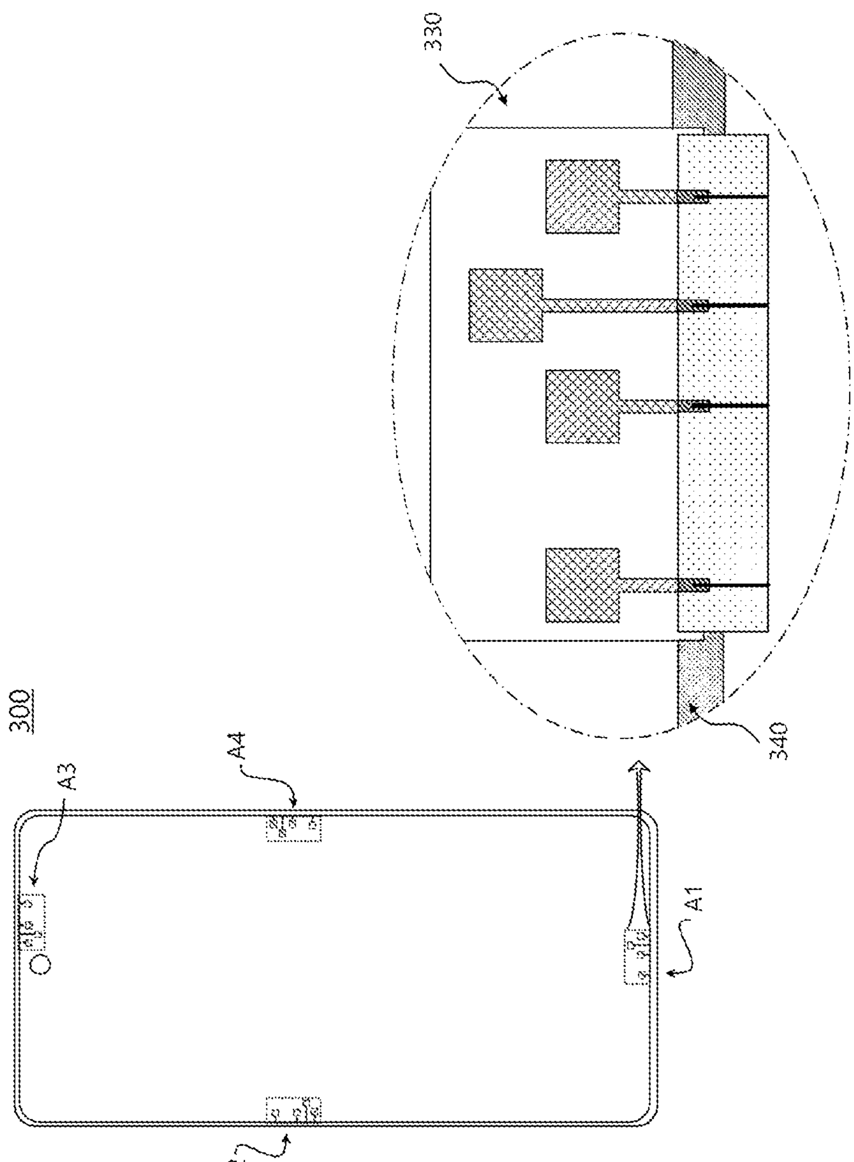
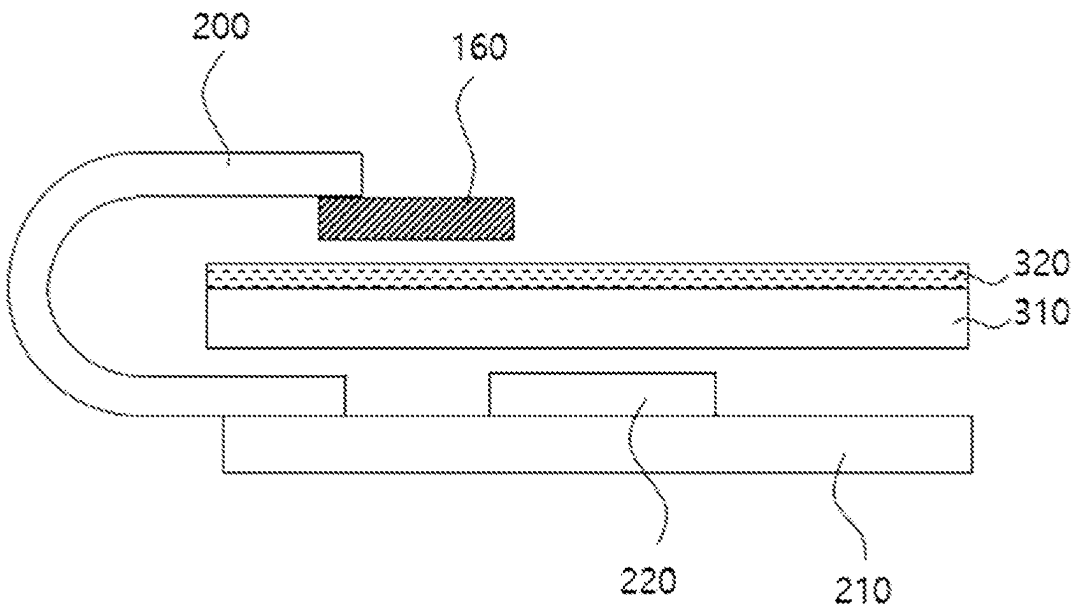


FIG. 8



ANTENNA STRUCTURE AND DISPLAY DEVICE INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION AND CLAIM OF PRIORITY

This application claims the benefit under 35 USC § 119 of Korean Patent Application No. 10-2022-0024938 filed on Feb. 25, 2022 in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Technical Field

The present invention relates to an antenna structure and a display device including the same. More particularly, the present invention relates to an antenna structure including a plurality of radiators and a display device including the same.

2. Description of the Related Art

As information technologies have been developed, a wireless communication technology such as Wi-Fi, Bluetooth, etc., or a non-contact type sensing technology such as a gesture sensing, a motion recognition, etc., may be employed or embedded in an image display device, an electronic device, an architecture, etc.

As mobile communication technologies has been recently developed, an antenna for performing, e.g., communications in high-frequency or ultra-high frequency band may be coupled to various mobile devices.

Specifically, a wireless communication technology is combined with a display device and implemented in the form of, e.g., a smart phone. In this case, an antenna may be coupled to the display device to perform a communication function.

As the display device in which the antenna is employed becomes thinner and lighter, a space for an occupation of the antenna may also be decreased. Accordingly, the antenna may be included on a display panel in the form of a film or patch to be accommodated in a limited space.

However, when the antenna is disposed on a display panel, a coaxial circuit for transmitting and receiving a signal or a power supply may not be easily constructed. Further, sensitivity may be degraded due to the separate coaxial feeding circuit, or space efficiency and aesthetic characteristics of a structure to which an antenna device is applied may also be degraded.

For example, Korean Published Patent Application No. 10-2014-0104965 discloses an antenna device including an antenna element and a ground element.

SUMMARY

According to an aspect of the present invention, there is provided an antenna structure having improved signaling efficiency and radiation reliability.

According to an aspect of the present invention, there is provided an image display device including the antenna structure.

(1) An antenna structure, including: a first radiator group including a plurality of first radiators arranged in a first direction; a second radiator group including a plurality of second radiators arranged in a second direction perpendicu-

lar to the first direction; first transmission lines connected to each of the first radiators at the same layer as that of the first radiators; and second transmission lines connected to each of the second radiators at the same layer as that of the second radiators.

(2) The antenna structure according to the above (1), wherein the first radiator group and the second radiator group are disposed at the same layer.

(3) The antenna structure according to the above (1), wherein the first radiator group and the second radiator group share one radiator.

(4) The antenna structure according to the above (1), wherein the number of first radiators and the number of second radiators are the same.

(5) The antenna structure according to the above (1), further including a third radiator spaced apart from the first radiator group and the second radiator group.

(6) The antenna structure according to the above (5), wherein the third radiator serves as a transmission radiator, and the first radiator group and the second radiator group serve as reception radiators.

(7) The antenna structure according to the above (1), further including a dielectric layer on which the first radiator group and the second radiator group are disposed, wherein the first direction is inclined by a first tilting angle with respect to a length direction of the dielectric layer, and the second direction is inclined by a second tilting angle with respect to the length direction of the dielectric layer.

(8) The antenna structure according to the above (7), wherein each of the first tilting angle and the second tilting angle is in a range from 30° to 60°.

(9) The antenna structure according to the above (7), wherein the first radiator group includes two first radiators, and the second radiator group includes two second radiators.

(10) The antenna structure according to the above (9), wherein a ratio of a length difference between the second transmission lines relative to a length difference between the first transmission lines is in a range from 0.8 to 1.2.

(11) The antenna structure according to the above (7), further including: a first signal pad electrically connected to one end portion of each of the first transmission lines; and a second signal pad electrically connected to one end portion of each of the second transmission lines.

(12) The antenna structure according to the above (11), wherein the first signal pad and the second signal pad are arranged to form a single row in a third direction, and the first direction and the second direction are inclined by the first tilting angle and the second tilting angle, respectively, with respect to the third direction.

(13) The antenna structure according to the above (11), further including: a pair of first ground pads spaced apart from the first signal pad and disposed with the first signal pad interposed therebetween; and a pair of ground pads spaced apart from the second signal pad and disposed with the second signal pad interposed therebetween.

(14) The antenna structure according to the above (1), wherein the first radiators and the second radiators have a mesh structure.

(15) The antenna structure of the above (14), further including a dummy mesh pattern spaced apart from the first radiators and the second radiators around the first radiators and the second radiators.

(16) The antenna structure according to the above (1), further including an antenna unit spaced apart from the first radiators and the second radiators, wherein the antenna unit has a resonance frequency different from that of the first radiators and the second radiators.

(17) A motion recognition sensor including the antenna structure according to the above-described embodiments.

(18) A radar sensor including the antenna structure the above-described embodiments.

(19) A display device, including: a display panel; and the antenna structure according to the above-described embodiments disposed on the display panel.

(20) The display device according to the above (19), wherein the first direction is inclined by a first tilting angle with respect to a length direction of the display panel, and the second direction is inclined by a second tilting angle with respect to the length direction of the display panel.

According to embodiments of the present invention, a first radiator group included in an antenna structure may include a plurality of radiators arranged in a first direction, and a second radiator group may include a plurality of radiators arranged in a second direction perpendicular to the first direction. Accordingly, a signal intensity of the radiators in the first direction and a signal intensity of the radiators in the second direction may be respectively sensed.

The first direction and the second direction may be inclined at a predetermined tilting angle with respect to one side of a dielectric layer or one side of a display panel. Accordingly, a length of the transmission line connected to each of first radiators may be relatively shortened, and thus signal transmission speed and efficiency may be improved. Further, the transmission line may be extended in a straight line without a bending area, so that signal loss and reduction may be prevented.

Further, a ratio of a length deviation between the second transmission lines to a length deviation between the first transmission lines may be adjusted within a predetermined range. Accordingly, a signal imbalance in the first and second directions may be prevented, and a motion or gesture sensing performance in all directions may be improved by the antenna structure.

The antenna structure may be electrically connected to a motion sensor circuit or a radar processor through a circuit board. Accordingly, a signal change by a sensing target may be transmitted to the motion sensor circuit or the radar processor, and a motion or a distance of the sensing target may be detected.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic plan view illustrating an antenna structure in accordance with exemplary embodiments.

FIGS. **2** and **3** are schematic plan views illustrating an antenna structure in accordance with exemplary embodiments.

FIGS. **4** and **5** are schematic plan views illustrating an antenna structure in accordance with exemplary embodiments.

FIG. **6** is a schematic plan view illustrating an antenna structure in accordance with exemplary embodiments.

FIGS. **7** and **8** are a schematic plan view and a schematic cross-sectional view illustrating an image display device in accordance with exemplary embodiments.

DETAILED DESCRIPTION

According to exemplary embodiments of the present invention, an antenna structure including a plurality of radiator groups arranged in different directions is provided.

Hereinafter, the present invention will be described in detail with reference to the accompanying drawings. However, those skilled in the art will appreciate that such embodiments described with reference to the accompanying drawings are provided to further understand the spirit of the present invention and do not limit subject matters to be protected as disclosed in the detailed description and appended claims.

The terms “first”, “second”, “third”, “one end”, “other end”, “top surface”, “bottom surface”, etc., are not intended to designate an absolute position or order, but relatively used to distinguish different elements or parts.

FIG. **1** is a schematic plan view illustrating an antenna structure in accordance with exemplary embodiments.

Referring to FIG. **1**, the antenna structure may include a dielectric layer **100**, and a first radiator group **110**, a second radiator group **120**, a first transmission line **114** and a second transmission line **124** formed on the dielectric layer **100**.

The dielectric layer **100** may include, e.g., a transparent resin material. For example, the dielectric layer **100** may include a polyester-based resin such as polyethylene terephthalate, polyethylene isophthalate, polyethylene naphthalate and polybutylene terephthalate; a cellulose-based resin such as diacetyl cellulose and triacetyl cellulose; a polycarbonate-based resin; an acrylic resin such as polymethyl (meth)acrylate and polyethyl (meth)acrylate; a styrene-based resin such as polystyrene and an acrylonitrile-styrene copolymer; a polyolefin-based resin such as polyethylene, polypropylene, a cycloolefin or polyolefin having a norbornene structure and an ethylene-propylene copolymer; a vinyl chloride-based resin; an amide-based resin such as nylon and an aromatic polyamide; an imide-based resin; a polyethersulfone-based resin; a sulfone-based resin; a polyether ether ketone-based resin; a polyphenylene sulfide resin; a vinyl alcohol-based resin; a vinylidene chloride-based resin; a vinyl butyral-based resin; an allylate-based resin; a polyoxymethylene-based resin; an epoxy-based resin; a urethane or acrylic urethane-based resin; a silicone-based resin, etc. These may be used alone or in a combination of two or more thereof.

In some embodiments, an adhesive film such as an optically clear adhesive (OCA), an optically clear resin (OCR), etc., may be included in the dielectric layer **100**.

In some embodiments, the dielectric layer **100** may include an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, glass, or the like.

In an embodiment, the dielectric layer **100** may be provided as a substantially single layer.

In an embodiment, the dielectric layer **105** may include a multi-layered structure of at least two or more layers. For example, the dielectric layer **100** may include a base layer and an antenna dielectric layer, and may include an adhesive layer between the base layer and the antenna dielectric layer.

Impedance or inductance may be formed by the dielectric layer **100**, so that a frequency band for operating or driving the antenna structure may be adjusted. In some embodiments, a dielectric constant of the dielectric layer **100** may be adjusted in a range from about 1.5 to 12. If the dielectric constant exceeds about 12, a driving frequency may be excessively reduced, and driving in a desired high frequency or ultra-high frequency band may not be implemented.

The first radiator group **110** may include a plurality of first radiators **112** arranged in a first direction. For example, the first radiators **112** may be spaced apart from each other along a first axis X1 extending in the first direction. The first axis X1 may be defined as an imaginary straight line passing through centers of the first radiators **112** and extending in the first direction.

The second radiator group **120** may include a plurality of second radiators **122** arranged in a second direction. For example, the second radiators **122** may be spaced apart from each other along a second axis X2 extending in the second direction. The second axis X2 may be defined as an imaginary straight line passing through centers of the second radiators **122** and extending in the second direction.

Accordingly, a signal intensity of the first radiator group **110** in the first direction and a signal intensity of the second radiator group **120** in the second direction may be changed according to a position of a sensing object.

Each of the first radiators **112** and each of the second radiators **122** may be driven independently from each other. Thus, the change in the signal intensity according to the position or distance of the sensing object may be measured in the first direction and the second direction, respectively, and thus motion, gesture and distance of the sensing object may be sensed.

In exemplary embodiments, the first direction and the second direction may perpendicularly cross each other. Accordingly, the antenna structure may sense the change of the signal intensity in the two orthogonal axes X1 and X2 and may transmit the detected change to a motion sensor driving circuit or a radar processor. Based on the collected information, the driving circuit or processor may measure distances or motions in all directions on, e.g., an X-Y coordinate system.

In some embodiments, the first radiators **112** may be arranged with a constant interval. For example, a spacing distance in the first direction between adjacent first radiators **112** may be the same as each other.

In some embodiments, the second radiators **122** may be arranged with a constant interval. For example, a spacing distance in the second direction between adjacent second radiators **122** may be the same as each other.

The signal intensity in the first direction or the second direction may be measured per a constant distance by the first radiators **112** and the second radiators **122** disposed with the constant interval. Accordingly, for example, the change of the signal intensity in the first direction or the second direction according to the change in the position of the sensing object may be more accurately measured.

In an embodiment, the spacing distance between the first radiators **112** in the first direction may be the same as the spacing distance between the second radiators **122** in the second direction.

In some embodiments, the first radiator group **110** and the second radiator group **120** may share one radiator **112** and **122**. For example, the first radiator group **110** and the second radiator group **120** may share a common radiator **112** and **122** disposed in an area where a first axis X1 and a second axis X2 intersect.

The common radiator **112** and **122** may serve as a reference point for measuring the change of the signal intensity in the first and second directions. For example, the change in the position of the sensing object may be sensed by measuring the changes of the signal intensities in the first axis X1 and the second axis X2 based on the signal intensity of the common radiator **112** and **122**.

In some embodiments, the number of first radiators **112** and the number of second radiators **122** may be the same. For example, the first radiator group **110** and the second radiator group **120** may include the same number of radiators. In this case, measurement of the signal change in the first direction and the signal change in the second direction may be balanced, and sensitivity and sensing performance in the first direction and the second direction may be improved together.

In some embodiments, each of the radiators **112** and **122** may be designed to have a resonance frequency of high or ultra-high frequency bands corresponding to, e.g., 3G, 4G, 5G or higher bands. For example, the resonance frequency of each of the radiators **112** and **122** may be about 50 GHz or more, specifically from 50 GHz to 80 GHz, and more specifically from 55 GHz to 77 GHz.

The antenna structure may include transmission lines connected to the radiators **112** and **122**. The transmission line may transfer a driving signal or a power from an antenna driving integrated circuit (IC) chip to the radiator, and may transfer an electromagnetic wave signal or electrical signal of the radiator to the antenna driving IC chip or the motion sensor driving circuit.

The transmission line may include a first transmission line **114** connected to the first radiator **112** and a second transmission line **124** connected to the second radiator **122**.

The first transmission line **114** may be disposed at the same layer as that of the first radiator **112**. For example, the first transmission line **114** may be integrally connected to the first radiator **112** and extend from one end of the first radiator **112**.

The second transmission line **124** may be disposed at the same layer as that of the second radiator **122**. For example, the second transmission line **124** may be integrally connected to the second radiator **122** and extend from one end of the second radiator **122**.

In an embodiment, the first transmission line **114** may be provided to each of the first radiators **112**, and the second transmission line **124** may be provided to each of the second radiators **122**.

In some embodiments, the first transmission line **114** and the second transmission line **124** may be disposed at the same level as that of the first radiator group **110** and the second radiator group **120** on the dielectric layer **100**. The transmission lines **114** and **124** may be arranged at the same level as that of the radiators **112** and **122**, so that an additional coaxial feeding for signal input/output and feeding may not be required. For example, an antenna on display (AOD) in which the antenna structure is placed on a display panel may be implemented.

In some embodiments, the antenna structure may further include a third radiator **132** spaced apart from the first radiator group **110** and the second radiator group **120**.

The third radiator **132** may serve as a transmission radiator for a motion sensing, and may radiate a radio wave toward the sensing object. The first radiator **112** and the second radiator **122** may serve as reception radiators, and may receive the radio wave reflected from the sensing target.

Accordingly, the antenna structure may receive and transmit wireless signals for the sensing object, and ae motion sensor may measure attenuation or increase of the signal based on a position change and a distance of the sensing object.

In an embodiment, the antenna structure may include a third transmission line **134** electrically connected to the third radiator **132**. The third transmission line **134** may be disposed at the same layer or level as that of the third radiator **132**.

In some embodiments, one end portions of the transmission lines **114**, **124** and **134** may be connected to the radiator **112**, **122** and **132**, and the other end portions of the transmission lines **114**, **124** and **134** may be bonded to a circuit board.

The circuit board may include, e.g., a flexible printed circuit board (FPCB). For example, a conductive bonding structure such as an anisotropic conductive film (ACF) may be bonded onto the other end portions of the transmission lines **114** and **124**, and then the circuit board may be pressed onto the conductive bonding structure.

The antenna driving IC chip may be mounted on the circuit board. In an embodiment, an intermediate circuit board such as a rigid printed circuit board may be disposed between the circuit board and the antenna driving IC chip. In an embodiment, the antenna driving IC chip may be directly mounted the a circuit board.

In an embodiment, the motion sensor driving circuit may be mounted on the circuit board. For example, the antenna structure and the circuit board are electrically connected, so that signal transmission/reception information of the antenna structure may be transferred to the motion sensor driving circuit. Accordingly, a motion recognition sensor including the antenna structure may be provided.

In some embodiments, a ground layer may be formed on a bottom surface of the dielectric layer **100**. Generation of an electric field in the transmission line may be more promoted through the ground layer, and an electrical noise around a feeding line may be absorbed or shielded.

In some embodiments, the ground layer may be included as an individual component of the antenna structure. In some embodiments, a conductive member of a display device in which the antenna structure is employed may serve as the ground layer.

The conductive member may include, e.g., various wires such as a gate electrode of a thin film transistor (TFT) included in a display panel, a scan line or a data line, or various electrodes such as a pixel electrode and a common electrode.

In an embodiment, a metallic member such as a SUS plate, a sensor member such as a digitizer, a heat dissipation sheet, etc., disposed at a rear portion of the image display device may serve as the ground layer.

FIG. **2** is a schematic plan view illustrating an antenna structure according to exemplary embodiments.

Referring to FIG. **2**, the first direction and the second direction may be inclined at a predetermined tilting angle with respect to a length direction or a width direction of the dielectric layer **100**.

The term "width direction" as used herein may refer to a horizontal direction of the dielectric layer **100** in FIGS. **1** to **5**, and may refer to, e.g., a third direction. The term "length direction" as used herein may mean a vertical direction perpendicular to the horizontal direction of the dielectric layer **100** in FIGS. **1** to **5**.

For example, the first direction may be inclined at a first tilting angle θ1 with respect to the length direction or the width direction of the dielectric layer, and the second direction may be inclined at a second tilting angle θ2 with respect to the length direction or the width direction of the dielectric layer.

The radiators **112** and **122** may be arranged to be oblique with respect to a lateral side of the dielectric layer **100** to reduce a length difference between the transmission lines **114** and **124** connected to the radiators **112** and **122**.

If the length difference between the transmission lines **114** and **124** connected to the radiators **112** and **122** becomes increased, a resistance and a signal line loss may be increased to deteriorate sensitivity and signal efficiency. Accordingly, an accuracy of a signal input to each of the radiators **112** and **122** and an electromagnetic wave signal emitted from each of the radiators **112** and **122** may be degraded, and a signal with respect to the sensing object may be inaccurately measured to cause measurement errors of the position and distance.

Further, if the length difference between the first transmission lines **114** and the length difference between the second transmission lines **124** are different from each other, a signal sensitivity in the first direction and a signal sensitivity in the second direction may become different. Thus, measurement of the change of the position change and distance along two axes may become inaccurate, and gesture and motion sensing performance may be degraded.

In exemplary embodiments, the first radiator group **110** and the second radiator group **120** may be disposed to be inclined with the predetermined tilting angle with respect to the length direction or the width direction of the dielectric layer **100**, so that the length of the transmission line may be reduced to prevent the increase of the signal loss and the resistance.

Additionally, the length difference between the transmission lines **114** and **124** connected to the radiators **112** and **122** may be reduced, so that the sensitivity difference in the first axis and the second axis may also be reduced. Thus, the change of the signal intensity based on the motion of the sensing object may be more accurately measured.

In some embodiments, the first transmission line **114** may extend in a straight line from the first radiator **112**. In some embodiments, the second transmission line **124** may extend in a straight line from the second radiator **122**.

The transmission lines **114** and **124** may extend in a straight line, so the signal loss and noise generation due to bent or folded portions may be prevented. Accordingly, signal transmission/reception efficiency and sensitivity of the radiator groups **110** and **120** may be improved, and the accuracy of the motion and gesture detection may be improved.

In some embodiments, the first tilting angle θ1 and the second tilting angle θ2 may be in a range from 15° to 75° or from 30° to 60°, respectively. Within the above range, the first radiator group **110** and the second radiator group **120** may be symmetrically disposed at the same plane, so that the length difference between the first transmission lines **114** and the length difference between the second transmission lines **124** may be reduced.

Preferably, the first tilting angle θ1 and the second tilting angle θ2 may be 45°.

In an embodiment, the first transmission line **114** and the second transmission line **124** may extend to be parallel to each other. For example, the first transmission lines **114** and the second transmission lines **124** each extend in straight lines from one side of the radiators, and may be arranged along the length direction or the width direction (the third direction) of the dielectric layer.

In this case, each of the first direction and the second direction may be inclined by the tilting angle with respect to an arrangement direction of the transmission lines **114** and **124**.

In some embodiments, each of the first radiator group **110** and the second radiator group **120** may include a plurality of radiators. For example, each radiator group may include two or three radiators.

Referring to FIG. **2**, the first radiator group **110** may include two first radiators **112**, and the second radiator group **120** may include two second radiators **122**.

In some embodiments, the first radiator group **110** and the second radiator group **120** may share one radiator. For example, the first radiator group **110** and the second radiator group **120** may each consist of three radiators.

Accordingly, an area occupied by the radiators **112** and **122** and the transmission lines **114** and **124** may be reduced. Thus, the antenna structure can be miniaturized and integrated while having improved signal transmission and reception efficiency and motion sensing performance.

In an embodiment, a ratio of the length difference d2 between the second transmission lines **124** relative to the length difference d1 between the first transmission lines **114** may be in a range from 0.8 to 1.2, or from 0.9 to 1.1.

Within the above range, signal sensitivities in the first and second directions may be maintained as being similar to each other, and measurement errors due to the difference of signal sensitivities in each direction may be reduced.

Preferably, the length difference d1 between the first transmission lines **114** and the length difference d2 between the second transmission lines **124** may be substantially the same.

The radiators **112** and **122** and the transmission lines **114** and **124** may include silver (Ag), gold (Au), copper (Cu), aluminum (Al), platinum (Pt), palladium (Pd), chromium (Cr), titanium (Ti), tungsten (W), niobium (Nb), tantalum (Ta), vanadium (V), iron (Fe), manganese (Mn), cobalt (Co), nickel (Ni), zinc (Zn), tin (Sn), molybdenum (Mo), calcium (Ca) or an alloy containing at least one of the metals. These may be used alone or in combination thereof.

In an embodiment, the radiators **112** and **122** and the transmission lines **114** and **124** may include silver (Ag) or a silver alloy (e.g., silver-palladium-copper (APC)), or copper (Cu) or a copper alloy (e.g., a copper-calcium (CuCa)) to implement a low resistance and a fine line width pattern.

In some embodiments, the radiators **112** and **122** and the transmission lines **114** and **124** may include a transparent conductive oxide such as indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide (ZnOx), indium zinc tin oxide (IZTO), etc.

In some embodiments, the radiators **112** and **122** and the transmission lines **114** and **124** may include a stacked structure of a transparent conductive oxide layer and a metal layer. For example, the radiators **112** and **122** and the transmission lines **114** and **124** may include a double-layered structure of a transparent conductive oxide layer-metal layer, or a triple-layered structure of a transparent conductive oxide layer-metal layer-transparent conductive oxide layer. In this case, flexible property may be improved by the metal layer, and a signal transmission speed may also be improved by a low resistance of the metal layer. Corrosive resistance and transparency may be improved by the transparent conductive oxide layer.

In an embodiment, the radiators **112** and **122** and the transmission lines **114** and **124** may include a metamaterial.

The radiators **112** and **122** and the transmission lines **114** and **124** may include a blackened portion, so that a reflectance at surfaces of the radiators **112** and **122** and the transmission lines **114** and **124** may be decreased to suppress a visual pattern recognition due to a light reflectance.

In an embodiment, a surface of the metal layer included in radiators **112** and **122** and the transmission lines **114** and **124** may be converted into a metal oxide or a metal sulfide to form a blackened layer. In an embodiment, a blackened layer such as a black material coating layer or a plating layer may be formed on the metal layer. The black material or plating layer may include silicon, carbon, copper, molybdenum, tin, chromium, molybdenum, nickel, cobalt, or an oxide, sulfide or alloy containing at least one therefrom.

A composition and a thickness of the blackened layer may be adjusted in consideration of a reflectance reduction effect and an antenna radiation property.

The antenna structure may further include a signal pad. For example, the signal pads may include a first signal pad **116** connected to a terminal end portion of the first transmission line **114** and a second signal pad **126** connected to a terminal end portion of the second transmission line **124**.

In one embodiment, the signal pads **116** and **126** may be provided as substantially integral members with the transmission lines **114** and **124**. For example, the distal ends of the transmission lines **114** and **124** may be provided as signal pads **116** and **126**.

In some embodiments, a ground pad may be disposed around the signal pads **116** and **126**. For example, a pair of first ground pads may be disposed to face each other with the first signal pad **116** interposed therebetween. For example, a pair of second ground pads may be disposed to face each other with the second signal pad **126** interposed therebetween. The ground pad may be electrically and physically separated from the transmission lines **114** and **124** and the signal pads **116** and **126**.

In some embodiments, the first signal pads **116** and the second signal pads **126** may be arranged in the width direction or the length direction of the dielectric layer **100**, and may be arranged, e.g., in the third direction.

For example, the first signal pads **116** and the second signal pads **126** may be spaced apart from each other along a third axis X3 extending in the third direction. The third axis X3 may be defined as an imaginary straight line passing through the centers of the signal pads and extending in the third direction.

The first radiator group **110** and the second radiator group **120** may each be inclined at a predetermined tilt angle with respect to the arrangement direction of the signal pads **116** and **126**.

In an embodiment, the circuit board may be bonded together on the signal pads **116** and **126** and the other end portions of and the transmission lines **114** and **124**. The ground pads may be arranged around the signal pads **116** and **126**, so that bonding stability of the circuit board can be further improved.

In some embodiments, the antenna structure may include a third signal pad **136** electrically connected to one end portion of the third transmission line **134**. In an embodiment, the third signal pad **136** may be provided as a substantially integral member with the third transmission line **134**. For example, a terminal end portion of the third transmission line **134** may be provided as the third signal pad **136**.

In an embodiment, the antenna structure may include a pair of third ground pads disposed to face each other with the third signal pad **136** interposed therebetween.

In an embodiment, the antenna structure may include only one third radiator **132**. For example, the one third radiator **132** may be provided for one first radiator group **110** and one second radiator group **120**.

In one embodiment, the antenna structure may include a plurality of third radiators **132**. For example, the plurality of third radiators **132** may be spaced apart from the first radiators **112** and the second radiators **122** around the first radiator group **110** and the second radiator group **120**.

FIG. **3** is a schematic plan view illustrating an antenna structure in accordance with exemplary embodiments.

Referring to FIG. **3**, the antenna structure may further include a dummy mesh pattern **150** disposed around the first radiator group **110** and the second radiator group **120**. For example, the dummy mesh pattern **150** may be electrically and physically separated from the radiators **112**, **122** and **132** and the transmission lines **114**, **124** and **134** by a separation region **155**.

For example, a conductive layer including the metal or alloy as described above may be formed on the dielectric layer **100**. A mesh structure may be formed while the conductive layer is etched along a profile of the radiators **112**, **122** and **132** and the transmission lines **114**, **124** and **134**. Accordingly, the dummy mesh pattern **150** spaced apart from the radiators **112**, **122** and **132** and the transmission lines **114**, **124** and **134** may be formed by the separation region **155**.

In some embodiments, the radiators **112**, **122** and **132** and the transmission lines **114**, **124** and **134** may include a mesh structure. Accordingly, transmittance of the antenna structure may be improved, and optical properties around the radiators **112**, **122** and **132** may become uniform by the dummy mesh pattern **150**. Thus, the antenna structure may be prevented from being visually recognized.

In an embodiment, the radiators **112** and **122** and the transmission lines **114** and **124** may entirely include the mesh structure. In an embodiment, at least a portion of the transmission lines **114** and **124** may include a solid structure for a feeding efficiency.

In an embodiment, the signal pad and the ground pad may be formed as a solid metal pattern to reduce a feeding resistance through the circuit board and prevent signal loss.

FIGS. **4** and **5** are schematic plan views of an antenna structure in accordance with exemplary embodiments.

Referring to FIGS. **4** and **5**, the first radiator group **110** may include three first radiators **112**, and the second radiator group **120** may include three second radiators **122**.

In some embodiments, the first radiator group **110** and the second radiator group **120** may share one radiator. In this case, the first radiator group **110** and the second radiator group **120** may entirely include five radiators.

Referring to FIG. **4**, the transmission line connected to the radiator disposed at an intersection of the first axis X1 and the second axis X2 may have the shortest length.

Referring to FIG. **5**, the transmission line connected to the radiator disposed at the intersection of the first axis X1 and the second axis X2 may have the longest length.

FIG. **6** is a schematic plan view of an antenna structure in accordance with exemplary embodiments.

Referring to FIG. **6**, the antenna structure may further include an antenna unit **140** spaced apart from the first radiator **112**, the second radiator **122** and the third radiator **132**.

The antenna unit **140** may have a resonance frequency different from that of the first radiator **112** and the second radiator **122**. Accordingly, a signal radiation for the motion detection and an electromagnetic wave radiation for a communication may be implemented together in one antenna structure.

The antenna unit **140** may be used for a mobile communication in a high frequency or an ultra-high frequency band, and may transmit and receive signals in, e.g., 3G, 4G, 5G or higher frequency bands. For example, the resonance frequency of the antenna unit **140** may be in a range from about 20 to 45 GHz.

In an embodiment, the antenna unit **140** may be disposed at the same layer or at the same level as that of the first radiator **112** and the second radiator **122** on the dielectric layer **100**.

The antenna unit **140** may include a fourth radiator **142** and a fourth transmission line **144** connected to the fourth radiator **142**. The fourth radiator **142** may have, e.g., a polygonal plate shape.

In some embodiments, a plurality of fourth transmission lines **144** may be connected to one fourth radiator **142**. Accordingly, a plurality of polarization directions may be substantially provided.

For example, each of the two fourth transmission lines **144** may be connected to two vertexes of a lower side of the fourth radiator **142**. In this case, a feeding may be performed in two substantially orthogonal directions to the fourth radiator **142** through each of the fourth transmission lines **144**. Accordingly, dual polarization properties may be implemented from one radiator **142**. For example, vertical radiation and horizontal radiation properties may be implemented together from the antenna unit **140**.

In an embodiment, the antenna unit **140** may include a fourth signal pad **146** connected to one end portion of the fourth transmission line **144**, and a plurality of fourth ground pads **148** facing each other with the fourth signal pad **146** interposed therebetween.

In one embodiment, two fourth ground pads **148** may be disposed between the fourth signal pads **146**. For example, two fourth ground pads **148** facing each other with the fourth signal pad **146** therebetween may be provided to each of the fourth signal pads **146**.

In an embodiment, one fourth ground pad **148** may be disposed between the fourth signal pads **146**. For example, the fourth signal pads **146** adjacent to each other may share one fourth ground pad **148**.

In some embodiments, the antenna unit **140** may be formed of the above-mentioned metal or alloy, or may include a transparent metal oxide.

In some embodiments, the antenna unit **140** may include a mesh structure to improve transmittance. For example, the fourth radiator **142** and the fourth transmission line **144** may include the mesh structure.

In an embodiment, the fourth radiator **142** and the fourth transmission line **144** may include a solid structure. For example, a lower portion of the fourth radiator **142** and the fourth transmission line **144** may have the solid structure. In this case, the solid portion of the antenna unit **140** may be located in a non-display area of the display device.

In an embodiment, the fourth signal pad **146** and the fourth ground pad **148** may include the solid structure to reduce a feeding resistance and improve noise absorption efficiency and horizontal radiation properties.

In some embodiments, the antenna unit **140** may be spaced d apart from the first radiator group **110** and the second radiator group **120** by a distance of $\lambda/2$ or more. λ may be a wavelength corresponding to the lowest frequency among resonance frequencies of the antenna unit **140**, the first radiator group **110** and the second radiator group **120**. For example, the resonance frequency of the antenna unit **140** may be a frequency corresponding to the wavelength.

For example, a separation distance between the fourth radiator **142** and the first radiator **112** and a separation distance between the fourth radiator **142** and the second radiator **122** may be $\lambda/2$ or more. The separation distance may indicate the shortest straight linear distance between two radiators.

In this case, a sufficient separation distance between radiators covering frequencies of different bands may be achieved to suppress signal disturbance and interference, and to prevent deterioration of the motion sensing performance and signal properties of the antenna structure.

FIG. **7** is a schematic plan view illustrating a display device in accordance with exemplary embodiments.

FIG. **7** shows a front portion or a window surface of a display device **300**. The front portion of the display device **300** may include a display area **330** and a non-display area **340**. The non-display area **340** may correspond to, e.g., a light-shielding portion or a bezel portion of an image display device.

The above-described antenna structure may be disposed toward the front portion of the display device **300**, and may be disposed on, e.g., a display panel.

In some embodiments, the above-described antenna structure may be attached on the display panel in the form of a film.

In an embodiment, the antenna structure may be formed throughout the display area **330** and the non-display area **340** of the display device **300**. In an embodiment, the radiators **112** and **122** may be at least partially superimposed over the display area **330**.

In some embodiments, the antenna structure may be located at a central portion of one side of the display device. For example, the front portion of the display device may include a first area A1, a second area A2, a third area A3 and a fourth area A4 located at the central portions of four sides of the display device.

The antenna structure may be disposed at the first region, the second region, the third region or the fourth region of the display device **300**, so that deterioration of the motion sensing performance at any one side may be prevented, and the motion, gesture or distance of the sensing object in all directions may be detected on the front portion.

FIG. **8** is a schematic cross-sectional view illustrating a display device in accordance with exemplary embodiments.

Referring to FIG. **8**, a display device **300** may include a display panel **310** and an antenna structure **160** disposed on the display panel **310**.

In exemplary embodiments, an optical layer **320** may be further included on the display panel **310**. For example, the optical layer **320** may be a polarization layer including a polarizer or a polarizing plate.

In an embodiment, a cover window may be disposed on the antenna structure **160**. The cover window may include, e.g., glass (e.g., an ultra-thin glass (UTG)) or a transparent resin film. Accordingly, an external impact applied to the antenna structure **160** may be reduced or absorbed.

For example, the antenna structure **160** may be disposed between the optical layer **320** and the cover window. In this case, the dielectric layer **100** and the optical layer **320** disposed under the radiators **112** and **122** may serve as dielectric layers of the radiators **112** and **122** together. Accordingly, an appropriate permittivity may be achieved so that the motion sensing performance of the antenna structure **160** may be sufficiently implemented.

For example, the optical layer **320** and the antenna structure **160** may be laminated using a first adhesive layer, and the antenna structure **160** and the cover window may be laminated using a second adhesive layer.

For example, a flexible printed circuit board **200** may be bent along a lateral curved profile of the display panel **310** to be disposed at a rear portion of the display device **300** and extend to an intermediate circuit board **210** on which the driving IC chip is mounted (e.g., the main board).

The flexible printed circuit board **200** and the intermediate circuit board **210** may be bonded or connected to each other using a connector, so that the feeding to the antenna structure **160** and the antenna driving control by the driving IC chip may be implemented.

In some embodiments, a motion sensor driving circuit **220** may be mounted on the intermediate circuit board **210**. In an embodiment, the motion sensor driving circuit **220** may be a proximity sensor, a gesture sensor, an acceleration sensor, a gyroscope sensor, a position sensor, a magnetic sensor, etc.

In some embodiments, the first radiator group **110** and the second radiator group **120** may be coupled to the motion sensor driving circuit **220**. For example, the antenna structure **160** may be electrically connected to the motion sensor driving circuit **220** through the flexible circuit board **200** connected to the intermediate circuit board **210**. Thus, signals sensed by the radiators **112** and **122** may be transmitted/provided to the motion sensor driving circuit **220**.

In an embodiment, the change of the signal intensities of the first radiator group **110** and the second radiator group **120** based on a movement of the sensing object mat be detected to measure a positional change of the sensing object. For example, the motion sensor driving circuit **220** coupled with the antenna structure **160** may detect a motion by detecting a signal change corresponding to a movement of the sensing object.

For example, the first radiator group **110** may detect the movement of the sensing object in the first direction. The second radiator group **120** may detect the movement of the sensing object in the second direction. Thus, the change of the signal intensity according to the motion/position in the first axis and the second axis may be provided from the antenna structure **160** to the motion sensor driving circuit **220**, and the motion and gesture along each axis may be measured in the motion sensor driving circuit **220**.

In one embodiment, the motion sensor driving circuit **220** may a include motion detection circuit. Signal information transmitted from the antenna structure **160** may be converted/calculated into a location information or a distance information through the motion detection circuit.

In an embodiment, the antenna structure **160** may be electrically connected to a radar sensor circuit, and thus signal transmission/reception information may be transmitted to a radar processor. For example, the antenna structure **160** may be connected to the radar processor through a circuit board. Accordingly, a radar sensor including the antenna structure may be provided.

The radar sensor may analyze a transmission/reception signal to detect information about the sensing object. For example, the distance to the sensing object may be measured by radiating radio waves from the antenna structure and receiving the radio waves reflected by the sensing object.

For example, the distance of the sensing object may be calculated by measuring a time required for a signal transmitted from the antenna structure to be reflected by the sensing object and then received by the antenna structure again.

What is claimed is:

1. An antenna structure comprising:

a first radiator group comprising a plurality of first radiators arranged in a first direction, the first radiators having the same resonance frequencies;

a second radiator group comprising a plurality of second radiators arranged in a second direction perpendicular to the first direction, the second radiators having the same resonance frequencies;

first transmission lines connected to each of the first radiators at the same layer as that of the first radiators; and

second transmission lines connected to each of the second radiators at the same layer as that of the second radiators; and

a third radiator spaced apart from the first radiator group and the second radiator group,

wherein the third radiator is disposed at only one side from the first radiator group of both sides from the first radiator group and the second radiator group,

the third radiator serves as a transmission radiator, and the first radiator group and the second radiator group serve as reception radiators, wherein the first radiator group and the second radiator group share one radiator, and

the shared radiator is disposed in an area where a first axis defined as an imaginary straight line passing through centers of the first radiators and extending in the first direction, and a second axis defined as an imaginary straight line passing through centers of the second radiators and extending in the second direction intersect each other.

2. The antenna structure according to claim **1**, wherein the first radiator group and the second radiator group are disposed at the same layer.

3. The antenna structure according to claim **1**, wherein the number of first radiators and the number of second radiators are the same.

4. The antenna structure according to claim **1**, further comprising a dielectric layer on which the first radiator group and the second radiator group are disposed,

wherein the first direction is inclined by a first tilting angle with respect to a length direction of the dielectric layer, and the second direction is inclined by a second tilting angle with respect to the length direction of the dielectric layer.

5. The antenna structure according to claim **4**, wherein each of the first tilting angle and the second tilting angle is in a range from 30° to 60°.

6. The antenna structure according to claim **4**, wherein the first radiator group comprises two first radiators, and the second radiator group comprises two second radiators.

7. The antenna structure according to claim **6**, wherein a ratio of a length difference between the second transmission lines relative to a length difference between the first transmission lines is in a range from 0.8 to 1.2.

8. The antenna structure according to claim **4**, further comprising:

a first signal pad electrically connected to one end portion of each of the first transmission lines; and

a second signal pad electrically connected to one end portion of each of the second transmission lines.

9. The antenna structure according to claim **8**, wherein the first signal pad and the second signal pad are arranged to form a single row in a third direction, and

the first direction and the second direction are inclined by the first tilting angle and the second tilting angle, respectively, with respect to the third direction.

10. The antenna structure according to claim **1**, wherein the first radiators and the second radiators have a mesh structure.

11. The antenna structure of claim **10**, further comprising a dummy mesh pattern spaced apart from the first radiators and the second radiators around the first radiators and the second radiators.

12. The antenna structure according to claim **1**, further comprising an antenna unit spaced apart from the first radiators and the second radiators,

wherein the antenna unit has a resonance frequency different from that of the first radiators and the second radiators.

13. A motion recognition sensor comprising the antenna structure according to claim **1**.

14. A radar sensor comprising the antenna structure according to claim **1**.

15. A display device, comprising:

a display panel; and

the antenna structure according to claim **1** disposed on the display panel.

16. The display device according to claim **15**, wherein the first direction is inclined by a first tilting angle with respect to a length direction of the display panel, and the second direction is inclined by a second tilting angle with respect to the length direction of the display panel.

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